

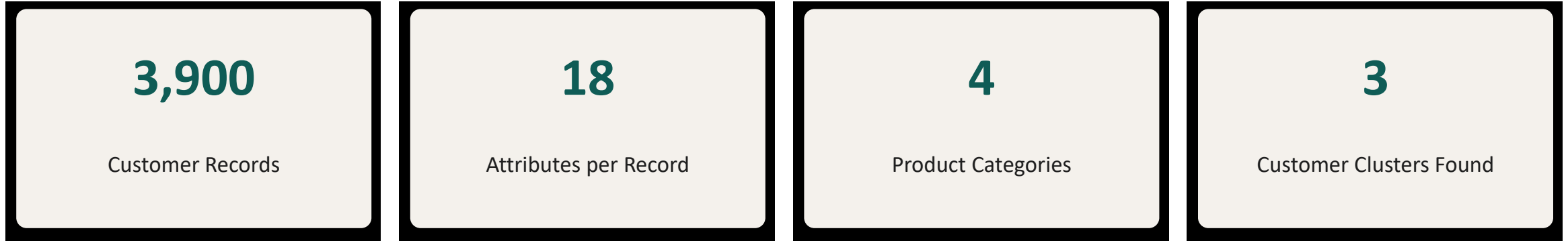
CUSTOMER SEGMENTATION

Shopping Trends Analysis · Project CDACL-002

Discount strategy · Spending patterns · K-Means clustering

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About the Dataset



Source: MySQL database (project_shopping_trends) — table shopping_table, verified row-for-row against duplicate table via NOT IN query.

Objective:

(1) Identify discount-worthy categories, (2) analyze spending by age/season/location, (3) cluster customers into 3+ groups for targeted marketing.

TASK 1

Which Categories Need Discounts?

IDENTIFY THE PRODUCT CATEGORIES WHERE DISCOUNTS SHOULD BE APPLIED

DISCOUNT ANALYSIS BY PRODUCT CATEGORY

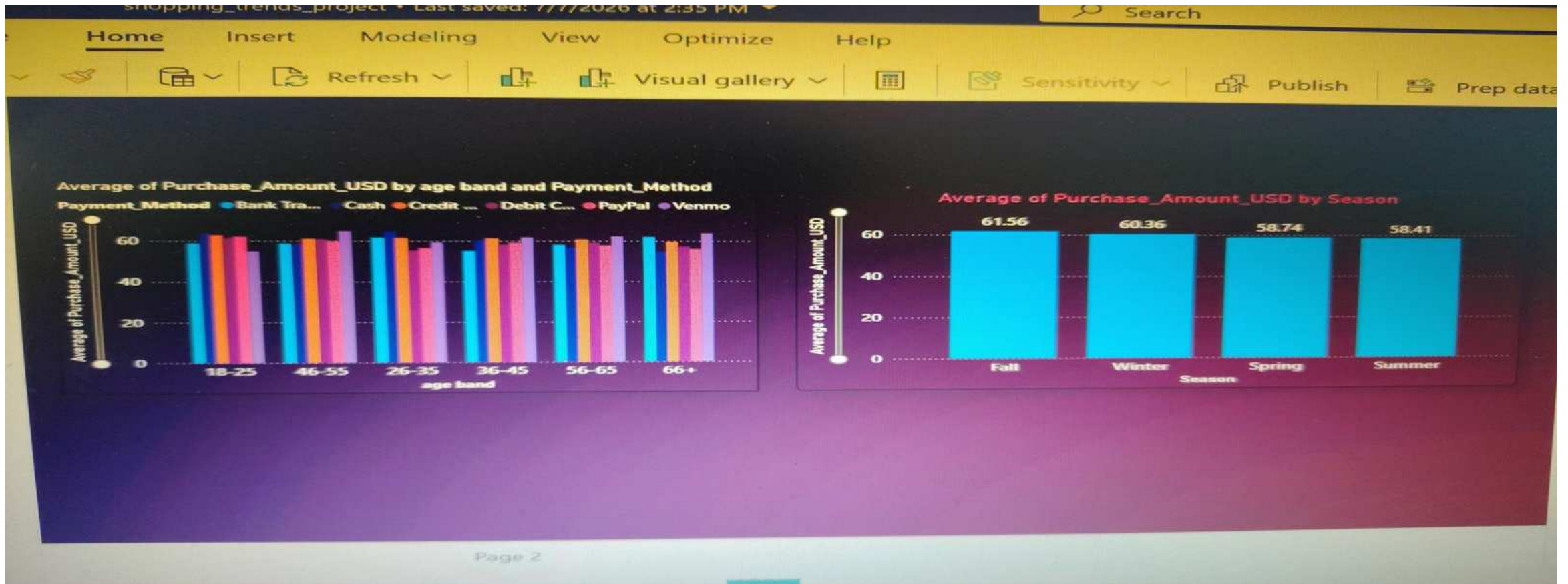
Category	Average of Purchase_Amount_USD	Average of Review_Rating	Count of Customer_ID	Discount %
Clothing	60.03	3.72	1737	42.08%
Footwear	60.26	3.79	599	43.24%
Accessories	59.84	3.77	1240	43.79%
Outerwear	57.17	3.75	324	44.44%
Total	59.76	3.75	3900	43.00%

Key Findings

- Purchase amount (\$57–60) and rating (3.72–3.79) are nearly identical across categories
- Volume varies sharply: Clothing (1,737) & Accessories (1,240) vs. Footwear (599) & Outerwear (324)
- Discount rates are already similar (42–44%)
— low volume isn't a discount problem
- Recommendation: seasonal, timed discounts for Footwear & Outerwear

TASK 2

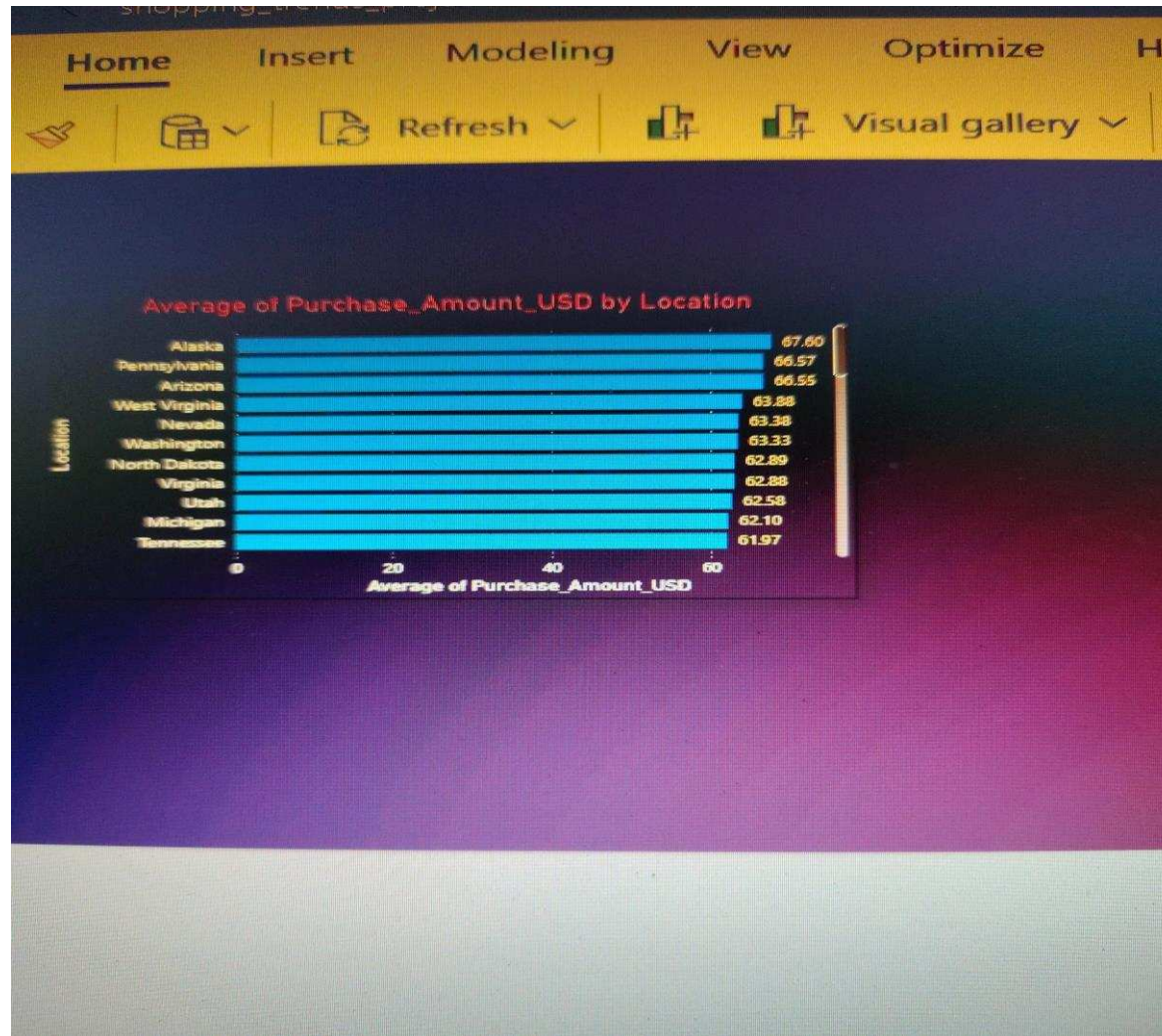
Spending by Age, Payment Method & Season



Spending is remarkably stable across age bands (~\$55–65) and seasons (~\$58–62). Venmo shows slightly higher spend among the youngest (18–25) and oldest (66+) groups; Fall is the highest-spending season, Summer the lowest — but the gap is small.

TASK 2

Spending by Location



Key Findings

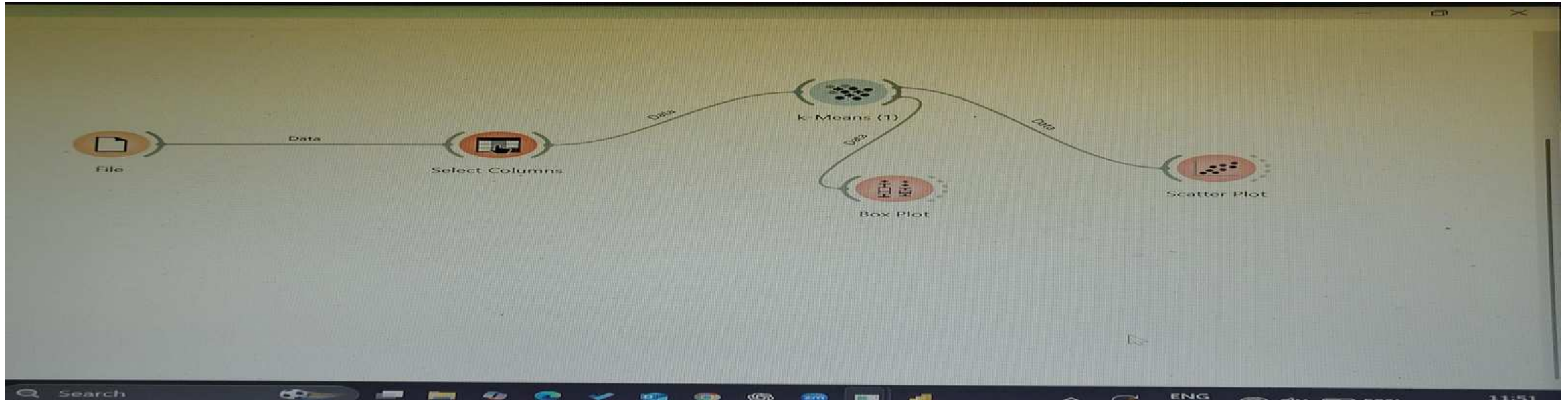
- Alaska (\$67.60), Pennsylvania (\$66.57) and Arizona (\$66.55) spend the most per purchase
- Lower end (Tennessee, Michigan) still averages ~\$62 — a gap of only ~\$6 top to bottom
- Location has a mild effect on spend, but not a dramatic one

Task 2 Takeaway

Purchase amount is driven more by product category than by who, when, or where a customer shops.

TASK 3

Clustering Methodology



1

Select Features

Age, Purchase Amount, Review Rating,
Previous Purchases

2

Preprocess

Normalize all features to a common
scale

3

K-Means (k=3)

KMeans++ init, 10 re-runs, 300 max
iterations

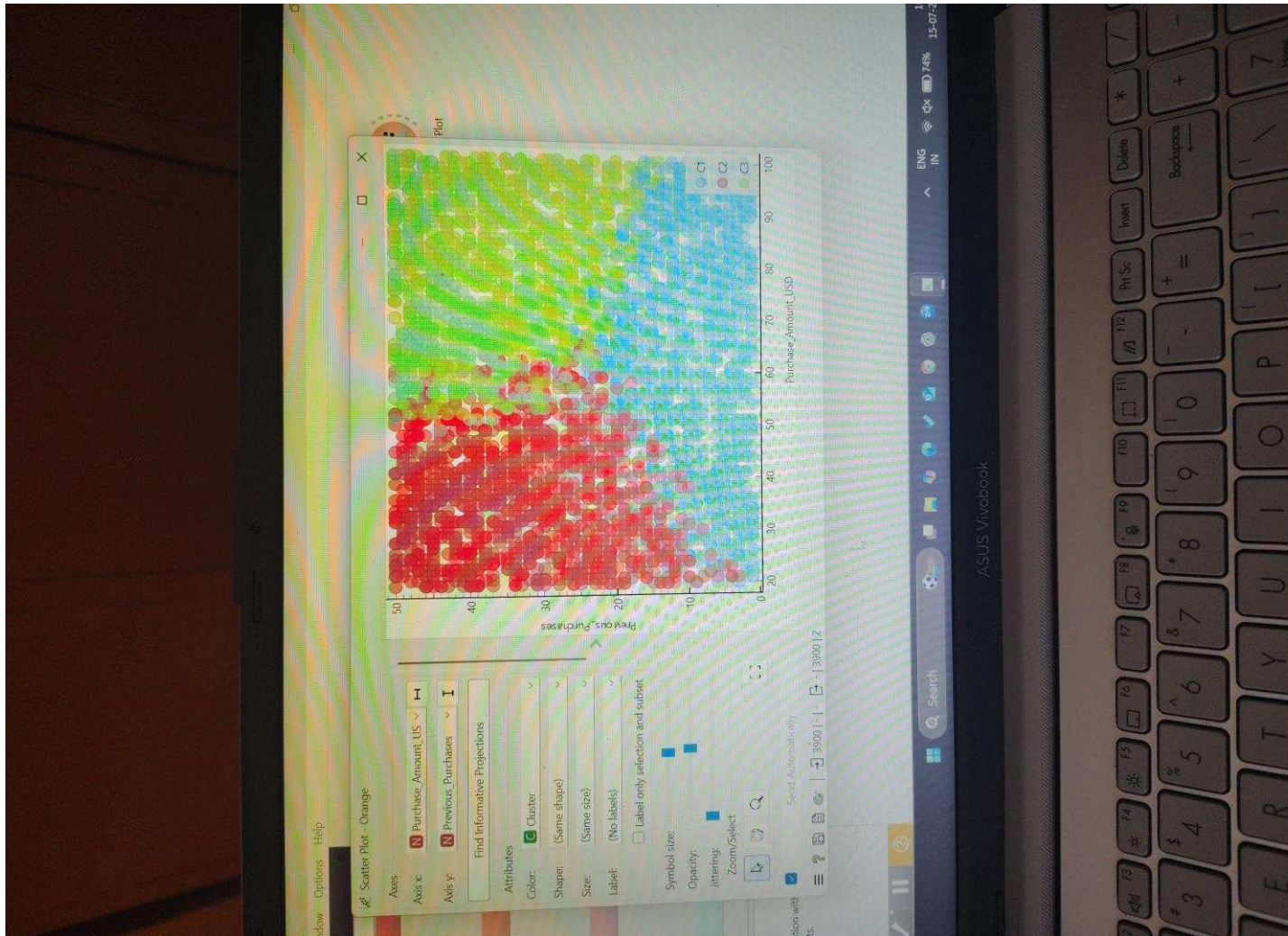
4

Visualize

Scatter Plot + Box Plots to profile each
cluster

TASK 3

Three Distinct Customer Clusters



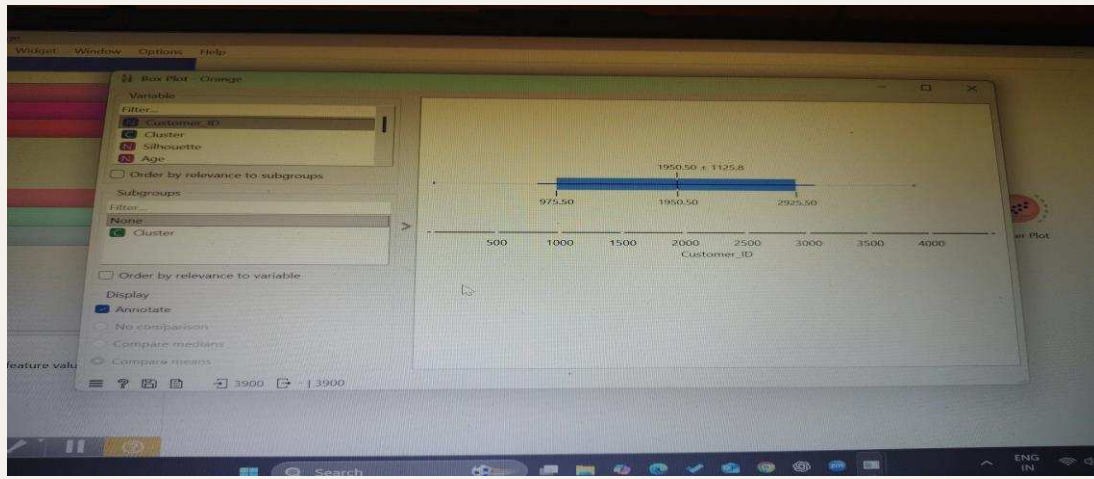
- **C1 — New High-Spenders**
Mid-high spend, low purchase history
- **C2 — Loyal Budget Shoppers**
High purchase history, low spend
- **C3 — VIP Loyal Customers**
High spend AND high purchase history

Purchase Amount vs. Previous Purchases cleanly separates all three groups, with almost no overlap between clusters.

TASK 3

Cluster Comparison — Box Plots

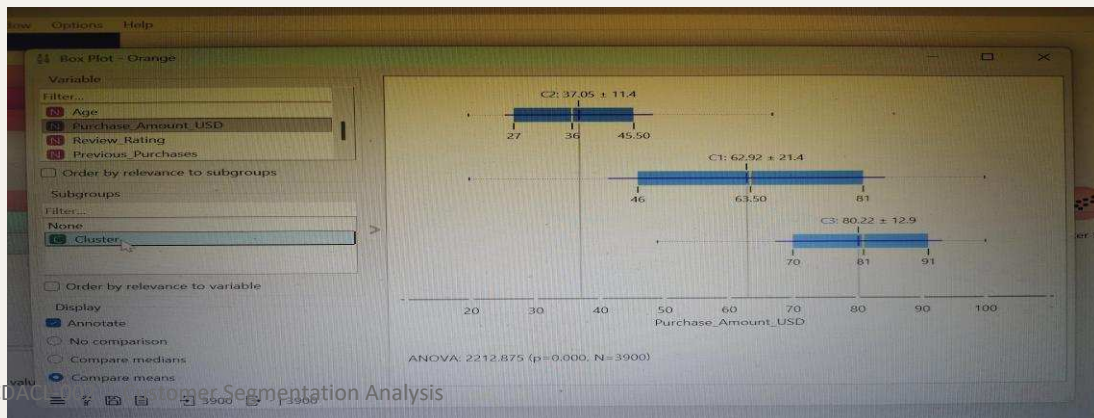
Purchase Amount (USD)



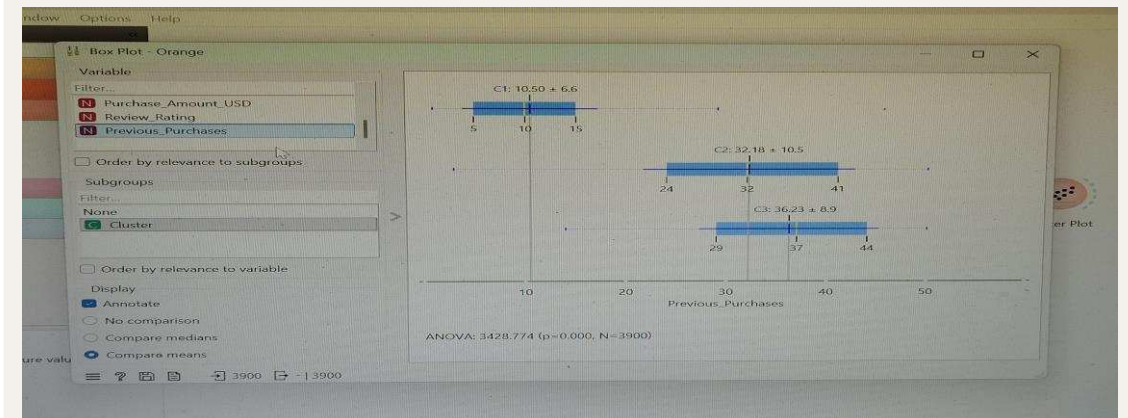
Age



Review Rating



Previous Purchases



TASK 3

Cluster Profiles at a Glance

Cluster	Avg. Purchase	Avg. Previous Purchases	Avg. Age	Avg. Rating
C1 — New High-Spenders	\$62.92	10.50	40.9	3.80
C2 — Loyal Budget Shoppers	\$37.05	32.18	46.6	3.59
C3 — VIP Loyal Customers	\$80.22	36.23	45.3	3.87

ANOVA testing confirms all four features differ significantly between clusters ($p = 0.000$). Purchase Amount and Previous Purchases show virtually no overlap between groups — the dominant factors. Age and Review Rating differ statistically but overlap heavily in practice — weak differentiators.

Dominant Factors

Purchase Amount (spend level) and Previous Purchases (loyalty/frequency) — not age or satisfaction — are what separate these customer groups.

TASK 3

Targeted Marketing Recommendations

C1 — New High-Spenders

RETAIN & CONVERT

Loyalty program enrollment, personalized follow-ups to build repeat-purchase behavior like Cluster 3.

C2 — Loyal Budget Shoppers

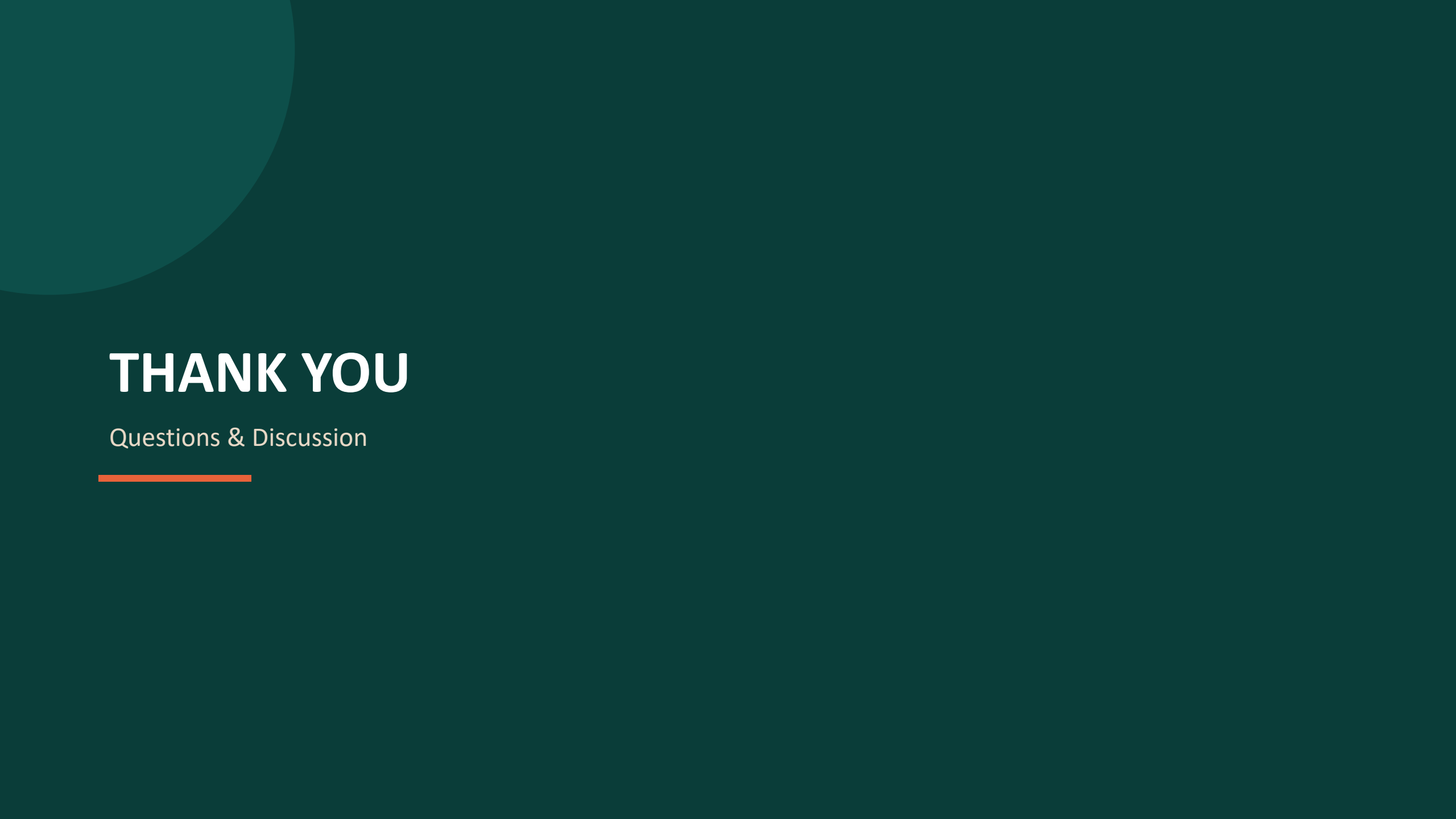
UPSELL & CROSS-SELL

Bundle deals and “frequently bought together” suggestions to raise average order value.

C3 — VIP Loyal Customers

REWARD & PROTECT

Exclusive early access, VIP perks, and referral incentives to safeguard this high-value segment.



THANK YOU

Questions & Discussion
